Module 8

Quick Reference and Course Summary

Learning objectives

- Provide a single quick reference and course summary for Verilog and SystemVerilog across IEEE versi…

Prerequisites

- See module README

Learning path

Learning Path

Diagram: Learning Path

(render with mmdc when available)

```
flowchart LR
T1["Version Timeline (One Table)"]
T1 --> N["Next module in course"]
```

Provide a single quick reference and course summary for Verilog and SystemVerilog across IEEE versions (1364-1995 through 1800-2017).

Overview

- This module is a **reference and summary** for the version-centric Verilog/SystemVerilog course (Mo…

How to learn this module

Suggested learning path (1/2)

- Skim this document — Goal, Overview, and Topics Covered set the IEEE scope for Module 8.
- Study design architecture — See how DUTs, examples, and testbenches fit together under `module8/`.
- Work through labs in order — Open `module8/EXAMPLES.md` and run each `make clean && make run` from…
- Run the full module script — `./scripts/module8.sh` from the repo root to simulate all examples and…
- Complete the exercises — Try each exercise in this document before reading solutions or peeking at…

Suggested learning path (2/2)

- Use as reference — Module 8 summarizes the full course; run examples to print quick-reference slice…
- Review Common Pitfalls — Can you explain each mistake and the fix?
- Self-check — Use `module8/CHECKLIST.md` before starting the next module.

Design architecture

1. Reference module layout (module8/)

- No new language — consolidates Modules 1–7 into quick-reference tables
- examples/ print version timelines, migration reminders, tool notes
- Use after completing the course or as a lookup during real projects
- scripts/module8.sh — runs reference printouts, not new RTL lessons

Rtl Architecture

```
Diagram: Rtl Architecture
(render with mmdc when available)
flowchart LR
M1["Module 1-3 Verilog"] --> M4["Module 4-6 SV design"]
M4 --> M7["Module 7 compare/migrate"]
M7 --> M8["Module 8 quick ref"]
M8 --> PROJ["Your project README standard"]
```

2. Course map architecture

- Module 1–3 — Verilog-only path (1995, 2001, 2005)
- Module 4–6 — SystemVerilog design path (2005, 2009/12, 2017)
- Module 7 — comparison and migration between standards
- Module 8 — this reference layer on top of the version stack

Verification Architecture

```
Diagram: Verification Architecture
(render with mmdc when available)
flowchart TB
REF["Module 8 tables"] --> PLAN["test plan: which moduleN.sh"]
PLAN --> RUN["run owning module scripts"]
M7T["Module 7 equiv tests"] --> RUN
```

3. How reference examples execute

- Most labs $display tables or cheat-sheet rows to stdout
- No new DUT hierarchy — output is documentation you can grep
- Pair with docs/MODULE8.md for full tables

Testing Methods

```
Diagram: Testing Methods
(render with mmdc when available)
flowchart TB
LOOKUP["construct_lookup"] --> REV["pick IEEE revision"]
REV --> SUB["pick design subset"]
SUB --> CI["CI flags match subset"]
CI --> REG["regression module1-7.sh"]
```

One-page course map

```
/*
 * Module 8: Course map - module number, title, content summary
 * Reference only; see module8/docs/course_map.md for links.
 */

module top;
    initial begin
        $display("=== Module 8: Course map (Modules 1-8) ===");
        $display("  1 | IEEE 1364-1995    | Verilog-95: wire/reg,
assign, always/initial, explicit sensitivity");
        $display("  2 | IEEE 1364-2001    | ANSI ports, always @*,
generate, signed, arrays, localparam");
        $display("  3 | IEEE 1364-2005    | Clarifications,
synthesizable subset, defparam deprecated");
        $display("  4 | IEEE 1800-2005    | logic,
always_comb/always_ff, interfaces, packages, unique/priority case");
        $display("  5 | IEEE 1800-2009/12 | Interface refinements,
```

Installation & execution

How to run this module

- From repo root: ./scripts/module8.sh --check — validate environment
- ./scripts/module8.sh --demo — print demo commands to run locally
- ./scripts/module8.sh --scaffold — create ~/unix_practice/ exercise files
- Open module8/EXAMPLES.md — run each numbered lab in your terminal
- Track progress with module8/CHECKLIST.md before moving on

Key files to study

Open these in the repo

- docs/MODULE8.md — master reference tables
- module8/examples/version_timeline/
- module8/examples/migration_steps/
- module8/examples/course_map/
- scripts/module8.sh

Verification & testing methods

1. Using Module 8 as closure

- Run ./scripts/module8.sh once to print all reference slices
- Cross-check “which standard has X?” against printed tables
- Bookmark slides.pdf and MODULE8.md for day-to-day lookup

2. When to re-run labs

- Re-run a specific Module 1–7 example when you need hands-on refresh
- Module 8 examples confirm you know where to find detail, not replace it

Version timeline printer

```
/*
 * Module 8: Version timeline - prints standard, year, role, module
 * Reference only; see docs/MODULE8.md and module8/docs/
 */

module top;
    initial begin
        $display("=== Module 8: Version timeline ===");
        $display("  1364-1995 | 1995 | First Verilog; base language
| Module 1");
        $display("  1364-2001 | 2001 | ANSI, @*, generate, signed
| Module 2");
        $display("  1364-2005 | 2005 | Last Verilog-only; defparam
deprecated  | Module 3");
        $display("  1800-2005 | 2005 | First SystemVerilog; logic,
always_comb/ff | Module 4");
        $display("  1800-2009 | 2009 | Interface refinements;
```

Syllabus topics

1. Version Timeline (One Table)

- See docs/MODULE8.md

Hands-on examples

Module 8 toolchain check

```
$ command -v iverilog >/dev/null && echo "Toolchain ready: $(iverilog
      -V 2>&1 | head -1)"
```

```
Terminal: module8_check
Toolchain ready: Icarus Verilog version 12.0 (stable) ()
```

Demo: One-page cheat sheet (version timeline, subsets, migration, ver...

```
$ cd module8/examples && make clean && make run
```

Terminal: ex01_quick_ref

```
rm -f top
iverilog -g2012 -o top quick_ref.sv
vvp top
=== Module 8: Quick Reference (one-page cheat sheet) ===
  Version timeline: 1364-1995 -> 1364-2001 -> 1364-2005 -> 1800-2005 -> 1800-2009 -> 1800-2012 -
  Verilog-only (1364): wire/reg, always @* or @(inputs), always @(posedge clk); no logic, no int
  SystemVerilog design (1800): logic, always_comb, always_ff, interface+modport, package+import,
  Migration 1364->1800: wire/reg->logic, @*->always_comb, posedge clk->always_ff, remove defpara
  Version selection: match tools/IP; state one revision (e.g. 1800-2017) and subset; document in
  Full tables and course map: module8/docs/ and docs/MODULE8.md
=== End quick reference ===
quick_ref.sv:16: $finish called at 0 (1s)
```

Demo: Version timeline (standard, year, role, module)

```
$ cd module8/examples/version_timeline && make clean && make run
```

Terminal: ex02_version_timeline

```
rm -f top
iverilog -g2012 -o top version_timeline.sv
vvp top
=== Module 8: Version timeline ===
  1364-1995 | 1995 | First Verilog; base language          | Module 1
  1364-2001 | 2001 | ANSI, @*, generate, signed            | Module 2
  1364-2005 | 2005 | Last Verilog-only; defparam deprecated  | Module 3
  1800-2005 | 2005 | First SystemVerilog; logic, always_comb/ff | Module 4
  1800-2009 | 2009 | Interface refinements; clarifications   | Module 5
  1800-2012 | 2012 | Checker; array/string methods          | Module 5
  1800-2017 | 2017 | Unified LRM; current standard          | Module 6
  Comparison and migration                                | Module 7
  Quick reference and summary                             | Module 8
  Path: 1->2->3->4->5->6 (version order); 7 (comparison); 8 (reference)
=== End version timeline ===
version_timeline.sv:20: $finish called at 0 (1s)
```

Demo: Verilog-only vs SystemVerilog design subset (ports, types, comb…

```
$ cd module8/examples/design_subset && make clean && make run
```

Terminal: ex03_design_subset

```
rm -f top
iverilog -g2012 -o top design_subset.sv
vvp top
=== Module 8: Design subset quick reference ===
  Verilog-only (1364):
    Ports: ANSI (2001+) or non-ANSI (1995)
    Types: wire, reg (no logic)
    Comb: always @* or always @(inputs)
    Seq:  always @(posedge clk) with <=
    No: logic, always_comb/always_ff, interfaces, packages, unique/priority case
  SystemVerilog design (1800):
    Ports: ANSI; logic typical
    Types: logic (single driver)
    Comb: always_comb
    Seq:  always_ff with <=
    Connectivity: interface+modport; package+import; unique/priority case
  Use 1364 when: tools or IP require Verilog only
  Use 1800 when: flow supports 1800; preferred for new RTL
=== End design subset ===
design_subset.sv:24: $finish called at 0 (1s)
```

Demo: Tool support summary (simulators, synthesis, lint, formal; Icar...

```
$ cd module8/examples/tool_support && make clean && make run
```

Terminal: ex04_tool_support

```
rm -f top
iverilog -g2012 -o top tool_support.sv
vvp top
=== Module 8: Tool support summary ===
  Simulators: 1364-2001/2005; 1800 subset varies by tool
  Synthesis:  Synthesizable subset of 1364 and/or 1800; revision varies
  Lint:       1364 and 1800; often selectable revision
  Formal:     Assertions/checkers (1800); revision varies
  Icarus:     Primarily 1364-2001/2005; limited SystemVerilog
  Verilator:  1364 and growing 1800 subset
  Commercial (VCS, Questa, Xcelium): Typically 1800-2005 through 1800-2017
  Match project standard to simulator, synthesis, and IP support.
=== End tool support ===
tool_support.sv:18: $finish called at 0 (1s)
```

Demo: "Which standard has X?" for key constructs (logic, always_comb,...

```
$ cd module8/examples/construct_lookup && make clean && make run
```

Terminal: ex05_construct_lookup

```
rm -f top
iverilog -g2012 -o top construct_lookup.sv
vvp top
=== Module 8: Construct lookup (first standard) ===
  logic              -> 1800-2005
  always_comb/always_ff -> 1800-2005
  interface, modport -> 1800-2005
  package, import    -> 1800-2005
  unique/priority case -> 1800-2005
  always @*          -> 1364-2001
  ANSI ports         -> 1364-2001
  generate           -> 1364-2001
  signed             -> 1364-2001
  localparam         -> 1364-2001
  defparam           -> 1364-1995 (deprecated in 1364-2005)
  Checker            -> 1800-2012
  Unified LRM        -> 1800-2017
  Full table: module8/docs/version_table.md
=== End construct lookup ===
construct_lookup.sv:24: $finish called at 0 (1s)
```

Demo: Course map (Modules 1–8 with title and content summary)

```
$ cd module8/examples/course_map && make clean && make run
```

Terminal: ex06_course_map

```
rm -f top
iverilog -g2012 -o top course_map.sv
vvp top
=== Module 8: Course map (Modules 1-8) ===
  1 | IEEE 1364-1995    | Verilog-95: wire/reg, assign, always/initial, explicit sensitivity
  2 | IEEE 1364-2001    | ANSI ports, always @*, generate, signed, arrays, localparam
  3 | IEEE 1364-2005    | Clarifications, synthesizable subset, defparam deprecated
  4 | IEEE 1800-2005    | logic, always_comb/always_ff, interfaces, packages, unique/priority ca
  5 | IEEE 1800-2009/12 | Interface refinements, checkers, array/string methods, assertions
  6 | IEEE 1800-2017    | Unified LRM, Verilog as subset of 1800, current standard
  7 | Version comparison| Side-by-side, migration patterns, tool support, checklists
  8 | Quick reference   | This module: tables, cheat sheet, course map
  Learning path: 1->2->3->4->5->6 (version order); then 7; use 8 as reference
  Links: module8/docs/course_map.md
=== End course map ===
course_map.sv:20: $finish called at 0 (1s)
```

Demo: Migration steps (1995→2001, 1364→1800, 1800→1800) and rules of…

```
$ cd module8/examples/migration_steps && make clean && make run
```

Terminal: ex07_migration_steps

```
rm -f top
iverilog -g2012 -o top migration_steps.sv
vvp top
=== Module 8: Migration steps quick reference ===
  1364-1995 -> 1364-2001:
    1. Ports -> ANSI  2. Comb -> always @*  3. generate/signed/localparam if needed  4. Test
  1364-2001/2005 -> 1800 (design subset):
    1. wire/reg -> logic  2. always @* -> always_comb  3. posedge clk -> always_ff
    4. Optional: interfaces, packages, unique/priority case  5. Remove defparam  6. Test
  1800-2005 -> 1800-2009/2012/2017:
    1. Run regression (usually compatible)  2. Adopt new features if supported  3. Set project s
  Rules: one block/feature at a time; keep external interface unchanged; one standard, document
  Full checklists: docs/MODULE7.md
=== End migration steps ===
migration_steps.sv:19: $finish called at 0 (1s)
```

Demo: Synthesizable do/avoid (assign, always_comb/@*, single driver;…

```
$ cd module8/examples/synthesizable_subset && make clean && make run
```

Terminal: ex08_synthesizable_subset

```
rm -f top
iverilog -g2012 -o top synthesizable_subset.sv
vvp top
=== Module 8: Synthesizable subset (both 1364 and 1800) ===
  Do: assign, always_comb or always @*, always_ff or always @(posedge clk)
  Do: generate, parameter, localparam, single driver per net
  Avoid: Delays (#), initial (except FPGA init per tool), fork/join
  Avoid: system tasks in RTL, defparam, multiple drivers (unless tri-state), unintended latches
  Check: Tool manual for exact synthesizable subset and revision
=== End synthesizable subset ===
synthesizable_subset.sv:15: $finish called at 0 (1s)
```

Demo: Version selection (match tools/IP, state one revision and subse...

```
$ cd module8/examples/version_selection && make clean && make run
```

Terminal: ex09_version_selection

```
rm -f top
iverilog -g2012 -o top version_selection.sv
vvp top
=== Module 8: Version selection quick reference ===
  Match: project standard to simulator, synthesis, and IP support
  State: one revision (e.g. IEEE 1800-2017) and subset (e.g. design only, no SVA in RTL)
  Document: in style guide or README; reference LRM and tool manuals
  Verilog-only: use when tools or IP require 1364; 1364-2005 as reference
  SystemVerilog design: use when flow supports 1800; 1800-2017 as reference
  Full checklist: docs/MODULE7.md (version-selection checklist)
=== End version selection ===
version_selection.sv:16: $finish called at 0 (1s)
```

Demo: Learning path (1→…→6, then 7; use 8 as reference) and where to…

```
$ cd module8/examples/learning_path && make clean && make run
```

Terminal: ex10_learning_path

```
rm -f top
iverilog -g2012 -o top learning_path.sv
vvp top
=== Module 8: Learning path and next steps ===
  Learning path: 1 -> 2 -> 3 -> 4 -> 5 -> 6 (version order)
  Then: Module 7 (comparison and migration); use Module 8 as reference anytime
  Where to go next:
    Project: Apply version selection and migration (Module 7); use Module 8 as quick reference
    LRM: IEEE 1800-2017 for authoritative syntax and semantics
    Tools: Simulator, synthesis, lint manuals for supported revision and synthesizable subset
    Related: RTL design patterns, UVM, SVA, tool-specific training
    New revisions: Add new module; update Module 7 and Module 8 tables
=== End learning path ===
learning_path.sv:18: $finish called at 0 (1s)
```

Demo: Common pitfalls quick reference (construct table, subset rule,...

```
$ cd module8/examples/pitfalls && make clean && make run
```

Terminal: ex11_pitfalls

```
rm -f top
iverilog -g2012 -o top pitfalls.sv
vvp top
=== Module 8: Common pitfalls (quick reference) ===
  1. Assuming a construct is in an older standard -> Use construct-by-standard table (Section 2)
  2. Mixing subsets without a rule -> Define one project subset and revision; document in style
  3. Skipping regression after migration -> Run sim/synth after every step; use Module 7 checkli
  4. Forgetting tool support -> Match project standard to tool-supported revision and synthesiza
  5. Treating Module 8 as replacement for 1-7 -> Use Module 8 for quick lookup; refer to Modules
  Full pitfalls: docs/MODULE8.md Common Pitfalls and How to Avoid Them
=== End pitfalls ===
pitfalls.sv:16: $finish called at 0 (1s)
```

Practice & assessment

What you should know

- ✓ Use the version timeline and construct-by-standard table to look up “which standard has X?”
- ✓ Use the design-subset quick reference to choose Verilog-only vs SystemVerilog design
- ✓ Use the migration quick reference and Module 7 checklists for migration and version selection
- ✓ Use the course map to find which module covers a given standard or topic
- ✓ Use this module as a single reference after completing the course

Exercises

- Lookup
- Subset
- Migration
- Course map
- Cheat sheet

Exercises

- Assuming a construct is in an older standard
- Mixing subsets without a rule
- Skipping regression after migration
- Forgetting tool support
- Treating this module as a replacement for Modules 1–7

Summary & next steps

- Provide a single quick reference and course summary for Verilog and SystemVerilog across IEEE versi…
- Complete module8/CHECKLIST.md
- Review module8/EXAMPLES.md and run each lab
- Next: Next module in course